SKOOLFINDER GHANA

SCHOOL OWNER & PROPRIETOR PORTAL

CONSOLIDATED MASTER BLUEPRINT v2.0

Complete product, commercial, UX, workflow, architecture, security and delivery specification

Baseline date: 4 November 2026
Status: Authoritative planning consolidation; implementation NOT authorised
References consolidated: School Owner Portal v1.0, v1.1, v1.2, v1.3, v1.4, v1.5, v1.6
Cross-portal dependency: Parents Portal v4.2, Theme B
Initial market: Greater Accra, Ghana
Development environment: Antigravity IDE

Document handling: Give this document together with the separately delivered Antigravity Master Implementation Prompt. The approved prices, quotas, six strategic decisions and public no-login school discovery rule are binding. Pending items are explicitly marked PENDING. No source code, migrations or deployments are authorised by this blueprint.

0. How to use this master blueprint

This is a detailed consolidation rather than a shortened executive summary. The requirements catalogue preserves the functional intent of every prior version, including later-phase functionality. It distinguishes (A) APPROVED business decisions, (D) product design requirements proposed for adoption, (P) pending commercial or operational policies, (F) future-release capabilities, and (V) technical facts that Antigravity must verify against the real repository. The document is a blueprint, not a claim that the software already exists.

- Read Sections 1-6 for authority, business model and school lifecycle.
- Read Sections 7-19 for every screen, function and operational module.
- Read Sections 20-29 for system architecture, security, automation and integrations.
- Read Sections 30-37 for phase-by-phase build, verification, testing, risks and handoff.
- Use the final traceability and decision register to prevent lower-priority historical wording from overriding later approvals.

1. Version lineage: details retained, not discarded

Version	Detailed contribution preserved	Authority
v1.0	Seven strategic pillars; 16 modules; school registration and approval; marketing; enquiries; admissions; student records; staff and recruitment; tenders and suppliers; school finance; communication; membership levels; owner and staff roles; analytics; security; future innovations; staged MVP; verification.	Source requirements unless superseded
v1.1	Six explicit decisions; one school per proprietor; monthly/annual; growth/ admissions first; Growth vacancy allowance; tender quotas plus paid posts; Greater Accra pilot; basic quota model; downgrade rules; Phase 2B idea.	Approved strategic rules
v1.2	Pricing research and launch price hypotheses, later approved in v1.5; annual discount; add-on hypotheses; three independent verification statuses; institution evidence; hybrid admissions form builder; detailed admin dashboard and automation catalogue.	v1.5 approval supersedes provisional prices
v1.3	12 sidebar groups; complete screen/ action design; standard CRUD and archive; parent enquiry inbox; form fields; tours; vacancy manager; tender states; future SIS and finance screens; staff role permissions; screen acceptance; edge cases.	Functional reference
v1.4	Conceptual data domains, source-of-truth decisions, application state machine, API domains, quota and credit sequencing, event contracts, RBAC + tenant + workflow checks, privacy, concurrency and security.	Architecture planning reference
v1.5	Explicitly APPROVED monthly/annual prices, membership feature/quota matrix, add-on charges, unapproved policy list and binding commercial rules.	Highest authority for commercials
v1.6	Audit-first governance; phases 0/1/2/2B/ 3/4/5; phase task IDs; 14 end-to-end tests; definition of done; go/no-go framework; development gates and separate implementation authorisation.	Highest authority for delivery sequencing

2. Final authority and change control

The controlling order is: explicit user-approved decisions in v1.1 and v1.5; this v2.0 resolution of those decisions; agreed Parents Portal v4.2 public-discovery and Theme B specifications; earlier detailed functional designs; implementation recommendations; actual codebase evidence. A verified implementation finding can identify a conflict, but cannot silently alter an approved business policy. Any change needs a decision record with rationale, impact, approval and effective date.

Status	Meaning	Handling
APPROVED	User explicitly selected or approved	Must not be overridden without a new user decision
DESIGN BASELINE	Fully specified recommended behaviour	Used for architecture and backlog; product-owner reviews implementation
PENDING	Not explicitly approved / missing legal or market detail	Do not hardcode or represent as final
FUTURE	Feature approved in roadmap but not available at launch	Design extensibility; do not sell as shipped
VERIFY	Depends on actual repository, providers or law	Inspect and cite evidence before making claims

3. Vision, positioning and strategic pillars

SkoolFinder Ghana is a connected school-discovery, growth and school-operations ecosystem. Parents discover and compare private schools; approved proprietors manage accurate listings and enquiries; admissions staff convert enquiries to applications; teachers and non-teaching candidates access vacancies; suppliers access tenders; later, school administrators run student, academic and fee workflows. The launch positioning is “helping schools get discovered, grow enrolment, manage operations and connect with the people and services they need.” A school’s subscription determines which released modules it can use, not whether its students’ data is protected.

Pillar	Capabilities / strategic purpose
1. Registration, verification and ownership	Register/claim school, assess authority and regulatory evidence, mediate disputes, create secure workspace.
2. Growth and visibility	Directory, public profile, search, Matchmaker, profiles, media, open days, enquiry sources, sponsored campaigns.
3. Student and academic operations	Future SIS, guardians, classes, attendance, assessments, reports, transfers and enrolment histories.
4. Recruitment and HR	Teaching/non-teaching adverts, applicant pipeline, interviews, staff records, employment workflows.
5. Procurement and suppliers	Tender/RFQ posting, supplier directory, bids, evaluations, awards, contracts.
6. School and platform finance	School invoices/fees and separate SaaS subscription/payment billing.
7. Intelligence, automation and operations	KPI dashboards, notices, reminders, audit, reporting, optional permission-aware AI.

4. Product ecosystem and cross-portal roles

Portal	Audience	Public/private data and interactions
Public Directory	Any visitor including unregistered	Browse/search/compare school details; authorised public listing only; never force login for discovery.

Portal	Audience	Public/private data and interactions
Parents Portal v4.2	Parents and guardians; guest browsing	Compare, Matchmaker, enquire, tours, applications, authenticated private tracking.
School Owner Portal	Owner, delegated school staff	Private school growth/operations, admissions, staffing, tenders, billing.
Teachers / Candidates	Teachers and non-teaching candidates	Public jobs, opt-in discoverability, scoped applications and notifications.
Suppliers Portal	Vendors and service providers	Supplier profile, public/invited tenders, confidential bid submissions.
Super Admin	Platform staff with limited roles	Institution approval, subscriptions, reports, moderation, disputes, escalations.
Shared services	Trusted backend only	Identity, schools, permissions, billing, events, files, audit, reporting.

The same human may be a parent, teacher and school staff member. Use one identity with explicitly scoped memberships/roles, not credential duplication or privilege inheritance. A single canonical school ID links all public and private records; a school profile is a moderated published view, not an independent school record.

5. Frozen strategic decisions from v1.1

ID	Approved decision	Precise consequence
DEC-01	One school per proprietor account initially	One active primary owner-school workspace per proprietor; delegated staff allowed; future multi-school-ready schema.
DEC-02	Monthly and annual billing	Separate price and renewal interval; annual discount approved later.
DEC-03	Admissions and growth before SIS	No full student/academic system in launch MVP; enrolment confirmation without SIS.
DEC-04	Limited Growth vacancies	Growth gets 2 simultaneously active slots per v1.5; higher tiers expanded.
DEC-05	Tender included quota plus additional paid posts	Quota accounting and purchase ledger with exact approved prices.
DEC-06	Greater Accra pilot then Ghana expansion	Region-scoped onboarding, Ghana-ready geography, public discovery not unnecessarily restricted.

6. FINAL APPROVED subscriptions, prices, entitlements and add-ons

6.1 Monthly and annual prices — APPROVED

Tier	Monthly (GH¢)	Annual (GH¢)	Annual saving (GH¢)
Free	0	0	-
Growth	99	990	198
Professional	249	2,490	498
Enterprise	499	4,990	998

Annual members receive 12 months for 10 months of standard price (16.666...% effective discount). These are final approved nominal prices but tax treatment, VAT/levy presentation, refund terms and payment processing charges remain pending verification. Growth is the first generally sold paid plan; Professional/ Enterprise functionality is not marketed as operational until released.

6.2 Exact tier quotas — APPROVED

Feature	Free	Growth	Professional	Enterprise
Schools per primary proprietor	1	1	1	1
Authorised user seats (owner counts)	1	3	10	25
Gallery images	5	20	50	100
Simultaneously active vacancies	0	2	10	30
Included published tenders per billing month	0	0	3	10
Basic parent enquiries	Yes	Yes	Yes	Yes
Full enquiry CRM	No	Yes	Yes	Yes
Admissions workflows	No	Yes	Yes	Yes
School tour bookings	No	Yes	Yes	Yes
Student information	No	No	Yes, after release	Yes, after release
Attendance and academics	No	No	Yes, after release	Yes, after release
School fee management	No	No	Yes, after release	Yes, after release
Advanced analytics	No	Limited	Yes, after release	Yes, after release
API integrations	No	No	Selected, after release	Expanded, after release

6.3 Add-on schedule — APPROVED

Add-on	Approved charge	Availability / unresolved conditions
Extra active vacancy slot	GH₵25	One slot valid for 30 days; renewal/ reposting details PENDING.
One tender publication	GH₵45	One publication; validity/expiry and refund policy PENDING.
Three tender publications	GH₵120	Bundle of three; credit validity PENDING.
Featured school promotion	Starting at GH₵75	Exact campaign sizes, placement/ duration PENDING; visibly label sponsored placements.
Additional authorised staff seat	GH₵10/month	Seat counting and annual add-on proration policy to detail.
SMS notifications	Usage-based	Exact unit rate/bundles PENDING; no invented cost.

6.4 Cross-cutting commercial controls

- Plan, price, subscription, billing interval, entitlement, usage counter, credit ledger, transaction, invoice, refund/adjustment and marketing campaign are distinct data concepts.
- Enforce entitlements on the server, along with user-role permissions and a feature-release flag. Avoid hardcoded frontend-only checks.
- Staff seat includes owner; seats may be expanded using approved add-on price.
- Active vacancy quota tracks simultaneous publications; drafts do not consume a slot; close/expiry releases under valid rules; reopening requires capacity.
- Tender allowance tracks NEW publications per monthly billing entitlement period, including annual subscribers; drafts do not consume a publication.
- Consume included monthly tender allowance before purchased credits; within purchased credits choose an expiry-aware order (design recommendation, not new commercial approval).
- Prevent two concurrent requests from consuming the same allowance; failed publication must not permanently consume a credit; repeated payment webhooks must not create duplicate credits.
- Subscription states: selected/pending, trial if configured, active, past due, grace, cancelled-at-term-end, expired, suspended; capture payment and entitlement snapshots.
- Downgrades preserve records, accessible billing history and permitted exports. Restrict disallowed premium writes; do not silently delete live vacancies, tenders, student files or histories.
- On renewal failure notify owner and apply a PENDING grace-period policy. Historical proposal was seven days, not approved.
- Free-tier parent enquiry receipt remains available; exact basic-response restrictions are PENDING. Do not discard valid enquiries simply because the school uses Free.
- Separate platform subscription revenues from future school-collected application fees and school fees, with distinct ledgers, receipts and reconciliation.
- Never treat initiated payment as success; verify provider response on backend; use idempotent payment processing. Confirm actual Mobile Money recurrence capability and provider fees before implementation.

7. School onboarding, ownership and verification: full specification

7.1 Three access journeys

Journey	Sequence	Safeguard
Register new school	Create/verify account > detect duplicates > complete institution wizard > upload evidence > submit > admin review > owner approval > workspace > profile publication	New entity only if no canonical match; separate approval from public publication.
Claim existing school	Find listing > submit authority evidence > check current owner > conflict resolution > review > attach to canonical record	Never create second school simply to bypass disputed ownership.
Join existing school as staff	Owner/admin invites email or account > identity confirmation > role assignment > accept invite > school membership	Invite expiry, seat count, permission scope and revocation enforced.

7.2 Detailed onboarding data dictionary

Data group	Required / possible fields	Review and constraints
Institution identity	Official name; operating name; registration details; date established; school type; description; logo	Protected legal identity changes require higher verification.
Ownership	Primary proprietor; representative; business/school authority; identity proof; official contact	Private documents; never expose national IDs publicly.
Location	Country; Greater Accra region; district; town; physical address; GPS pin; contact directions	Geolocation sourced/confirmed; multi-campus reserved.
Education	Nursery/KG/Primary/JHS/SHS etc.; curricula; grades; intake ages; day/boarding; gender policy; language	Education-level-sensitive regulatory evidence rules.
Operating details	Opening hours; school calendar; contact phones; official email; website	Verified contact and publication controls.
Regulatory evidence	Appropriate registration/licensing documents; issue/expiry; authority; file and review notes	NaSIA/other applicable regulators depend on educational level; no unsupported automatic accreditation claim.
Commercial	Free or paid plan; invoice contact; monthly/annual interval; billing history	Payment activation independent of school owner verification.
Media	Logo; facilities photos; video; captions; usage rights; publication status	Quota, moderation, privacy and permissions.

7.3 Full registration and approval states

Status	Allowed next steps	Owner-facing experience
Draft	Complete, submit, abandon	Resume wizard and checklist

Status	Allowed next steps	Owner-facing experience
Submitted	Review, request information	Review receipt and reference
Under Review	Approve, reject, escalate, request information	Status and outstanding requirements
Information Required	Resubmit, withdraw	Specific evidence requests and deadline
Approved Owner	Activate authorised workspace	Guided dashboard and onboarding
Rejected	Appeal or new corrected request	Reason and documented route
Suspended	Admin review, remediation, appeal	Restricted access and support instructions
Archived	Retained according to policy	No active management workspace

Three independent statuses must be maintained: authority-to-manage school, regulatory evidence/licensing status and subscription status. The public badge “School Profile Verified by SkoolFinder” may mean only the defined SkoolFinder checks, not official government accreditation. Store documentary evidence and source/verification date; renewal intervals must be document-specific, not one universal hardcoded rule.

7.4 Owner and school constraints

- One active primary proprietor per school and one primary school per proprietor at launch; no second workspace via hidden endpoint.
- A school with Nursery, KG, Primary and JHS can be one workspace if legally/operationally one institution; independently registered campuses need future classification policy.
- Multi-school later: model ownership using memberships and separate organisation concepts rather than literal one-to-one table architecture.
- Ownership dispute: freeze sensitive transfer, hold evidence securely, escalate to authorised verification reviewer, issue reasoned outcome and appeal path.
- Transfers retain school_id, approved public profile, admissions history, billing, audit and institution continuity; revoke the old owner in a controlled transaction.
- Approval of account, purchase of plan and publication of school profile are separate events; verified owner may have a draft unpublished profile.

8. Public school profile, SEO/discovery and marketing

8.1 Public profile fields and editorial workflow

- Name, brand logo, school description, mission/vision, contact information, approved map location and operating hours.
- Educational levels, grades/age ranges, curriculum, day/boarding, facilities, fees or optional ranges, school transport, extracurriculars, programmes, achievement descriptions, admissions criteria and current intake.
- Media gallery with photograph rights, captions/alt text, quota checks and careful child-image publication permissions.
- Verified licensing information only with properly defined label/status; never unsubstantiated “government approved” badges.

- Profile completeness indicator with actionable missing fields, not a misleading education quality rating.
- Draft > preview > submit > moderated/published public version > edit history; sensitive legal changes trigger re-verification.
- Profile appears consistently in Directory, Parents Portal, comparison, Matchmaker, SEO listing and any map view; public endpoint serves approved version only.
- Sponsored placements must be conspicuously identified, separate from organic suitability and independently reported.

8.2 Marketing modules and actions

Module	Screens / data	Actions
Visibility analytics	Visitors, profile views, source and timeframe	Filter, compare, inspect source, export permitted aggregates.
Open days/events	Event date, capacity, venue, description, registered attendees	Create, preview, publish, cancel, reschedule, close, record attendance.
Announcements	Public notices, school calendar items, approved campaigns	Draft, schedule, publish, unpublish, archive.
Sponsored campaigns	Placement, schedule, budget/price, purchase, reporting	Configure, preview labelled ad, pay, view impressions and outcomes.
School tours	Availability, visit type, slots, parent booking, reminders	Confirm, reschedule, cancel, register attendance, link enquiry.
Marketing dashboard	Enquiries by source, tour attendance, application conversion	View drilldown, date filtering and baseline comparisons.

9. Dashboard, navigation and UI system in full

9.1 Global application shell

Persistent element	Required interaction
School logo and school identity	Display authorised school, profile completeness and verification state.
Desktop left navigation	Role/plan/release-sensitive groups; preserve clear hierarchy.
Mobile navigation	Accessible drawer/bottom or compact menu; no loss of essential tasks.
Global search	Search ONLY permitted records and modules; no cross-school leaks.
Notification bell	Unread count; scoped details, destination, read/unread history.
My Tasks	Assigned and overdue tasks; open/in progress/waiting/completed/cancelled.
Subscription indicator	Current plan, renewal and warning; open billing center.
User menu	Account, security, MFA, support, sign out and context.
Breadcrumbs	Clear module and detail hierarchy.

Persistent element	Required interaction
Quick-create	Only permitted released actions: enquiry, tour, job, tender, etc.
Help	Context-specific guidance and support tickets.

9.2 Complete 12-group sidebar inventory preserved from v1.3

Navigation group	Included screens/subpages	Release
Dashboard	Overview; My Tasks; Notifications; Activity History	Launch
My School	School Information; Public Profile; Educational Levels; Facilities & Curriculum; Gallery & Media; Verification Documents; Profile Preview	Launch
Enquiries	All Enquiries; New Enquiries; Assigned to Me; Follow-ups; Closed Enquiries; Enquiry Analytics	Launch
Admissions	Admissions Overview; Applications; Application Form Builder; Intake & Capacity; Interviews & School Tours; Offers & Waiting List; Admissions Reports	Launch
Marketing	Visibility Dashboard; Events & Open Days; School Announcements; Sponsored Campaigns; Promotion History	Launch / expansion
Recruitment & HR	Job Vacancies; Applicants; Interviews; Staff Directory; Staff Documents; HR Administration	2B / 4
Tenders & Procurement	My Tenders; Create Tender; Supplier Bids; Tender Evaluation; Supplier Directory; Contracts & Purchases	2B / 4
Students & Academics	Student Directory; Student Profiles; Guardians; Classes & Streams; Attendance; Assessments & Results; Promotion & Transfers	Phase 3
School Finance	Finance Overview; Fee Structures; Invoices; Payments & Receipts; Balances; Reports & Reconciliation	Phase 3
Reports & Intelligence	Admissions Reports; School Visibility; Recruitment Reports; Tender Reports; Operational Analytics; Data Exports	Progressive
Subscription & Billing	My Membership; Compare Plans; Usage & Quotas; Renewals & Add-ons; Invoices & Transactions	Launch
Settings & Support	Staff & Permissions; Academic Configuration; Notifications; Account Security; Privacy & Exports; Help & Support; Audit History	Launch / progressive

Historical v1.3 uses the approximate phrase “60 core screens” and different counts for list/detail and shared pages. This is NOT a definitive route count; the exact route inventory must be reconciled from the

above and detailed modules below. Shared components, tabs and detail views should not be double-counted. Enterprise menu visibility does not imply unreleased modules are available.

9.3 Reusable interface templates

Template	Required behaviour
Overview dashboard	KPI cards, dates, drilldowns, task queue, activity log, chart or fallback table.
List / table	Search, filter, sort, paginate, bulk actions where authorised, empty/error/loading states.
Record detail	Summary, tabs, related records, activity, status and role-specific action bar.
Create/edit	Labels, validation, field help, save draft, cancel and confirmation.
Approval review	Evidence, reviewer notes, reason codes, approve/request/reject/escalate, audit.
Pipeline / board	Counts and transitions; authoritative status, accessible list alternative.
Documents	Private upload, file-type/size validation, preview, expiry and signed access.
Calendar	Availability and timezone, capacity, conflict handling, reschedule/cancel.
Reports	Date and school-level filters, chart, export, clear metric definitions.
Subscription	Plan comparison, usage, invoices, renewal, upgrade, add-on purchase.
Settings	Permissions, warnings, security settings, audit and change confirmation.

9.4 Global buttons and action contract

Control	Required effect, permission and safety
Add / Create	Open validated form; check role, release and plan before commit.
View	Open scoped record details, never infer access from ID alone.
Edit	Load version and permitted fields; detect concurrent modification.
Save Draft	Persist incomplete work without submission/public visibility or quota use.
Submit	Validate required fields; transition workflow and issue receipt.
Approve / Reject / Decline	Authorised role, reason, confirmation and audit; notify appropriate party.
Publish / Unpublish	Check readiness, moderation and credits; maintain publication history.

Control	Required effect, permission and safety
Pause / Resume / Reopen	Enforce state and active-slot rules.
Assign / Reassign	Validate assignee school membership and capability.
Archive / Delete	Archive consequential records; hard delete only through policy-driven legal procedure.
Duplicate	Create new draft without reusing sensitive identifiers or quota.
Search / Filter / Sort	Server-side scoped queries, pagination, accessible controls.
Import / Export	Validate data; school/field scope, logs, safe download and retention.
View History	Read status changes with actor/timestamp/reason.
Notify / Reply	Record communication with delivery state and consent/prefs.
Buy Add-on / Upgrade	Show price/terms, initiate provider, verify payment, grant once.

9.5 Dashboard overview — DASH-001

Cards: new enquiries, active admissions, tours scheduled, pending actions, school profile score, current plan, relevant vacancies/tenders when released. Charts: enquiry activity, sources, admissions by stage, conversion across steps. Activities: latest applications, staff assignments, document expiry, payment warnings. Quick actions: View Enquiries, Review Applications, Schedule Tour, Edit School Profile. Every count must have a defined school scope/timeframe and click through to filtered records; no hardcoded demo data in production. Personalised queues vary by role. Empty/loading/error states and accessible chart tables required.

10. Enquiry CRM in full

10.1 Public intake and list

Parents may browse and submit a basic enquiry without being forced to sign in. Validate legitimate contact method, prevent spam/abuse, record submitted fields, school ID, source (profile, Matchmaker, comparison, campaign or direct link), time and consent/notice. Prevent duplicate flooding without dropping legitimate follow-ups. Free-tier school receives the enquiry; advanced CRM tools are paid.

List column	Definition
Reference	Stable enquiry ID
Parent/contact	Permitted contact display
Interested level	Level/grade requested
Source	Campaign, directory, Matchmaker, compare, other
Received	Created timestamp
Owner	Assigned school user
Status	New/Contacted/Qualified/Tour Scheduled/Converted/Closed
Last activity	Latest reply/action

List column	Definition
Actions	View/Reply/Assign/Schedule/Close

10.2 Enquiry detail and interaction

- Show verified parent contact and message, requested level, source, complete conversation timeline, internal notes (not parent-visible), assigned staff, follow-up date, related tour and application.
- Reply via approved email/in-platform path with saved message, template support and delivery history.
- Assign only to authorised active staff; reassign logged.
- Schedule follow-up with owner, due date and reminder; detect duplicates.
- Schedule tour with slot capacity; link to enquiry.
- Convert to an application by initiating an applicant-authorised application, NOT copying data as an unconsented completed admissions form.
- Close with reason (enrolled elsewhere, uninterested, duplicate etc.), preserve history and allow authorised reopen.
- Reports: volume by source, response time, enquiry-to-tour, enquiry-to-application, enquiry-to-enrolment; make aggregation privacy-safe.

11. Admissions: complete design and state machine

11.1 Seven main functional screens and actions

Screen	Fields, controls and actions
ADM-001 Admissions Overview	KPI/pipeline, filters year/intake/level, due decisions, staff work queue; drill into lists.
ADM-002 Applications List	Applicant reference, grade, intake, status, assigned staff, dates; search/filter/sort, open, export by role.
ADM-003 Application Detail	Applicant/guardian/education/documents/timeline; request more info, schedule, offer, decline, waitlist, history.
ADM-004 Form Builder	Versioned standard fields + custom types; conditional logic; draft, preview, publish, close, duplicate.
ADM-005 Intakes & Capacity	Academic year, class level, opening/closing dates, seats, offers, acceptances and enrolments.
ADM-006 Tours & Interviews	Calendar/list, availability, capacity, confirmations, reminders, reschedule, cancel, attendance.
ADM-007 Offers & Waitlist	Draft/issue offers, expiry, parent acceptance, waitlist position and authorised enrolment confirmation.

11.2 Complete admission form sections and field types

Section	Fields and privacy
Applicant identity	Name, date of birth, intended level, intended year/intake, relevant identifiers.

Section	Fields and privacy
Parent / guardians	Contact, guardian relationship, communication preferences; verify authority for private tracking.
Previous education	Prior school, current level, transfer context and supported learning needs where legitimate.
Admission preferences	Day/boarding, curriculum, year, relevant school selection.
Documents	Reports/certificates etc. only where justified; private upload with access restrictions.
Declarations	Accuracy, privacy notice, permissions where required, terms of application.
Custom fields	Short/long text, number, date, radio, multi-select, yes/no, upload, conditional rules, declarations.

- Start from platform-standard form; school adds justified fields under controlled policy.
- Each form version is immutable for existing submissions; retain field IDs and original answers when new version published.
- Autosave drafts for authenticated applicants; guest application requires secure verification/resume method; do not require login for general school discovery.
- Intake opening and closing dates and capacity enforced; conditional fields and required documents validated.
- Submission creates a school-scoped application and parent-facing tracking record; acknowledgement and staff notification.
- A school may request additional documents/info and resume review; preserve timeline.
- No autonomous AI rejection or high-stakes decision; authorised human makes offers/declines.
- Application fees and refunds require a separately approved commercial/policy framework; never infer permitted amounts.
- Offer acceptance requires verified parent/guardian authority; enrolment confirmation does not depend on Phase 3 SIS.
- Future SIS creates or links a student with idempotent mapping and duplicate detection.

11.3 Detailed admissions states and transitions

From	To	Actor + prerequisites
Draft	Submitted	Parent or authorised representative; valid form/intake and contact checks.
Submitted	Under Review	Admissions officer/authorised staff.
Under Review	Information Required	Admissions officer; specific request and deadline.
Information Required	Under Review	Parent submits update; staff or configured review acknowledges.
Under Review	Assessment Scheduled	Admissions staff; valid event slot.
Assessment Scheduled	Decision Pending	Assessment completed as school policy requires.

From	To	Actor + prerequisites
Under Review	Decision Pending	Required checks complete; no assessment required.
Decision Pending	Offered	Authorised decision-maker; capacity and offer terms verified.
Decision Pending	Declined	Authorised decision-maker; recorded decision/reason.
Under Review / Decision Pending	Waitlisted	Admissions role; queue and eligible intake.
Waitlisted	Offered	Decision-maker; place available and waitlist policy.
Offered	Accepted	Verified parent/guardian within offer validity.
Offered	Declined by Parent	Verified representative; record response.
Accepted	Enrolled	Registrar/authorised school official; avoid duplicate enrolment.
Permitted active states	Withdrawn	Verified parent/guardian; record notice.
Offered	Expired	Automated expiry only under published offer terms.
Any correctable state	Reopened / corrected	Authorised admin with recorded reason; limited policies.

Do not allow arbitrary direct status updates. Before transition verify current status and version, school, staff role, plan availability, required documents/reviews, capacity, offer terms, idempotency, audit and notification. Enrolled is distinct from accepted and from Phase 3 creation of a student record. Waitlist position and capacity require consistent transactional logic.

11.4 Student-capacity logic and tour scheduling

- Capacity by school/level/class/academic intake; distinguish places advertised, offers outstanding, accepted offers and confirmed enrolments.
- When intake full, close applications or enable waitlist according to school-configured and approved policy.
- Tour types: personal visit, open day, interview/assessment; location, staff, timeslot, capacity, booking confirmation, reminders, cancellation and attendance.
- Prevent overbooking with server-side reservation, not only calendar display.
- Rescheduling preserves history and notifies participants; cancellation releases capacity under rules.

12. Recruitment, teaching and non-teaching, and HR

12.1 Teaching and non-teaching scope

Job categories include classroom/subject teachers, nursery/KG practitioners, teaching assistants, heads of department/principals, learning-support roles, administrators, office reception, finance/accounting, IT, cleaners, security, drivers, catering, librarians, maintenance and other relevant school staff. Vacancy publishing must not assume every applicant is a registered teacher. Candidate applications are school-scoped and require privacy safeguards.

12.2 Vacancy screens and form

Screen	Details
JOB-001 Vacancy overview	Active/draft/closed lists, quotas, search, status, source and candidate counts.
JOB-002 Create/Edit Vacancy	Title, category, department, level, employment type, job description, qualifications, experience, salary info if disclosed, job site, deadline, required docs, contact method.
JOB-003 Vacancy detail	Preview, edit, publish, pause, reopen, close, duplicate, archive, view history.
JOB-004 Applicants	Candidate submissions, documents and screening status; simple intake in Phase 2B, full ATS in Phase 4.
JOB-005 Job Analytics	Views/applications/sources, quota usage, time open, conversion (where available).

Vacancy action	Rules
Create / save draft	No active slot used; protected by role.
Publish	Check approved release, plan active limit or extra 30-day slot, required fields, moderation.
Edit	Validate sensitive changes, preserve publication revision.
Pause	Hide new applications; define quota handling and duration.
Close / expire	Stop accepting applicants and release active allowance under agreed rule.
Reopen	Check available slot; event logged; check applicable paid entitlement.
Duplicate / repost	New draft; avoid duplicate-ad loophole and carry no implicit free allowance.
Purchase slot	GH¢25, valid 30 days, backend payment verification and usage record.

12.3 Recruitment state machines and HR expansion

Vacancy lifecycle: Draft > Pending Review (if required) > Published/Open > Paused or Closed > Archived. Applicant lifecycle: Submitted > Screening > Shortlisted > Interview > Offered > Hired or Declined, plus Withdrawn. A job may be closed while applicants remain under review. Later HR records: staff directory, department, school role, qualifications, certification reminders, contracts, leave requests, onboarding, training, attendance and access revocation. Hiring is NOT automatic platform access provisioning. Payroll, biometric attendance and complex appraisals are deferred until compliance and demand validation.

13. Tenders, sourcing and procurement

13.1 Procurement categories and tender form

Potential school purchases: books, uniforms, furniture, stationery, laptops/ICT, laboratories, building works, maintenance, cleaning, security, catering, transport, ICT systems and professional services. Public supplier directory and informal RFQ support should coexist with formal tenders.

Tender form block	Fields / rules
Identity	Title, reference number, procurement type and category.
Description / specifications	Goods/services/works, units, quantities, deliverables and attached technical documents.
Supplier eligibility	Required registrations, capability, experience, qualifications and evidence.
Timeline	Publish, clarification, submission, opening/award dates as applicable.
Location	Delivery address/service location and timing.
Evaluation	Criteria, weights, approval authority, conflicts of interest and opening procedure.
Visibility	Public or invite-only with permitted supplier list.
Communication	Clarification questions, updates, contact and recorded answers.

13.2 Tender screens and lifecycle

Screen	Main functions
TEN-001 My Tenders	Draft/published/closed/awarded lists; search/filter/sort; quota balances.
TEN-002 Create Tender	Validated multi-section form; save, preview, edit, submit/ publish.
TEN-003 Supplier Bids	Receive confidential files and prices; view only according to opening state.
TEN-004 Evaluation	Document and score qualified bids; keep reviewer notes and audit; later phase.
TEN-005 Supplier Directory	Search approved vendors, shortlist, RFQ and optional invitations.
TEN-006 Contracts / Purchases	Award decisions, contracts, deliveries and milestone history; later phase.

Tender states: Draft > Awaiting Publication > Published/Open > Submission Closed > Evaluation > Awarded or Cancelled > Archived. Tender amendments require visible versioning and supplier notifications. Late bids are rejected or handled under a documented tender-specific policy. For sealed tenders, evaluation access must be blocked until authorised opening. Commercial tender-credit purchase is unrelated to and must never influence procurement award.

13.3 Tender commercial and security rules

- Included monthly tender allowance: Free 0; Growth 0; Professional 3; Enterprise 10. Growth can buy tender publications.
- Add-ons: GH¢45 per single publication; three for GH¢120. Terms/credit validity PENDING.
- Drafts do not consume credits; publishing consumes exactly one entitlement under atomic reservation/commit.
- Use included quota first, then valid purchased credits (recommendation to confirm in operational policy).

- Reposting expired tender generally requires new entitlement; corrections to existing tender do not necessarily consume new credit.
- Tender cancellation/refund depends on PENDING published terms; never silently credit or debit without audit.
- Confidential bidder documents restricted to originating school and authorised procurement roles, with extra protection for sealed bid windows.
- Evaluation, conflicts of interest, award and contract management deferred to Phase 4; simple response intake can launch Phase 2B.

14. Student Information System: future Phase 3 with full requirements

No Student Information System will be delivered as part of the initial Growth/admissions MVP. However, its logical boundaries and intake-to-student migration are designed now. All student data is private by default; sensitive safeguarding and health content requires minimum collection and restricted access.

Screen / module	Fields and functions
SIS-001 Student Directory	Unique student ID, status, class, filters; add/import, view, permitted export, archive.
SIS-002 Student Profile	Name, enrolled school, level, guardians, enrolment/status history and restricted documents.
SIS-003 Guardian Relationships	Multiple verified guardians, authority and contact preferences, linking/unlinking with audit.
SIS-004 Classes & Streams	Create grade/stream, assign student, class capacity and staff assignment.
SIS-005 Attendance	Present/absent/late, daily teacher marking, corrections, reports and notifications.
SIS-006 Assessments & Results	Subjects, grading rubrics, assignments, teacher mark entry, review/approval, report cards.
SIS-007 Promotion & Transfers	Year progression, repeating, class change, transfers, alumni, withdrawal histories.
SIS-008 Student Documents	Enrolment docs and limited sensitive attachments, private storage and audit.

14.1 Student lifecycle and operational rules

Applicant > admitted > offered/accepted > enrolment-confirmed > active student (when SIS active) > transferred, withdrawn, graduated or alumnus. Avoid silently deleting students; use status histories and controlled archival. Imports require field mapping, preview, row validation, dedupe and partial failure reporting. Guardian access must be verified by school, not inferred from same surname/email. Teachers see assigned classes and permitted records only. Corrections to attendance/results carry actor, reason and history. Student and child data must not feed marketing targeting or public directory analytics.

14.2 Extended management modules

- Timetable; lessons/homework; exams; learning reports; staff attendance and training; library assets; transport and meal management; inventory, facilities and maintenance are expansion options.

- Sensitive disciplinary/safeguarding casework requires dedicated policy, strict need-to-know access and oversight; not an ordinary note field available to all teachers.
- Parents Portal future child-facing views: attendance summaries, authorised academic reports, parent-school notices and appointments; per-child relationship verification required.
- Group management and multi-campus HQ remain future features; no multi-school feature at launch even on Enterprise plan.

15. School finance vs SaaS billing

FIN screen	Functions and controls
FIN-001 Finance Overview	Collections, outstanding balances, date/term filters and account reconciliation.
FIN-002 Fee Structure	Charges by grade/class/term, versioning, discounts/scholarship rules.
FIN-003 Invoices	Issue student invoices, instalments, due dates and status.
FIN-004 Payments / Receipts	Record cash/bank/verified online transfers; allocate to invoices; numbered receipts.
FIN-005 Outstanding Balances	Balance after payments, adjustments, credits and refunds; search/report.
FIN-006 Financial Reporting	Collections, ageing, reconciliation and permission-controlled exports.
FIN-007 Refund / Adjustment	Delegated approvals, reversal records, immutable payment references and audit.

No school fee payment functionality in initial MVP unless separately approved. School subscriptions and their GH¢ charges are platform billing; school fees belong to each school's accounting domain. Payment processing uses approved providers rather than directly storing card details or accepting custody of school funds without appropriate arrangements. A receipt is never silently edited after issuance; use reversals/adjustments. Fees, application charges, taxes and refunds must be explicitly configured and reviewed before enabling.

16. Communications, notifications and tasks

16.1 Communication channels and consent

Use in-app notifications and email as core channels; SMS can be enabled as separately priced usage; WhatsApp is an optional authorised integration. Messages include admissions, enquiries, visits, job applications, supplier responses, fees (future) and school announcements. School staff must not message all SkoolFinder parents as a marketing list without appropriate permissions and consent.

16.2 Notification and automation catalogue: v1.2 preserved

ID	Trigger	Proposed action
AUT-01	New school registration	Notify verification reviewers.
AUT-02	Missing verification evidence	Send targeted completion reminder.

ID	Trigger	Proposed action
AUT-03	School ownership approved	Create onboarding tasks and send owner confirmation.
AUT-04	New parent enquiry	Notify assigned school admissions staff.
AUT-05	Enquiry unanswered	Schedule follow-up reminder under configured SLA.
AUT-06	Admissions application submitted	Acknowledge parent; notify school.
AUT-07	Admissions information requested	Secure notification to applicant.
AUT-08	School tour scheduled	Confirmation and time-based reminder.
AUT-09	Admissions decision issued	Notify verified parent/guardian.
AUT-10	Vacancy deadline approaching	Notify recruitment staff.
AUT-11	Tender deadline approaching	Notify procurement team and permitted bidders.
AUT-12	Tender quota exhausted	Explain paid-credit or upgrade options.
AUT-13	Subscription renewal approaching	Send billing reminder.
AUT-14	Payment confirmed	Verify and reconcile, activate entitlement exactly once.
AUT-15	Subscription past due	Begin approved grace/notification policy.
AUT-16	Verification evidence nearing expiry	Notify school and reviewers for reassessment.

All notifications need event ID, school/record scope, template version, delivery status, retry/duplicate control, language/accessibility consideration, preference handling and audit. n8n may orchestrate reminders and integrations, but it cannot be the authoritative service for billing, permissions or admission decisions. Time-bound jobs must support retries and observability.

17. Roles, RBAC, school access and staff administration

Role	Main powers	Sensitive restrictions
Primary Proprietor	School workspace governance, subscription, authorised role delegation, broad reporting	No bypass of platform compliance, student privacy or dual-approval policies.
School Administrator	Delegated school operations/ configuration	Cannot transfer primary ownership without approved process.
Headteacher / Principal	Academic and school operational oversight	Access only authorised categories.
Admissions Officer	Enquiries, applicants, tours, admissions decisions as delegated	No unrelated school finance or procurement bids.
Registrar	Enrolment and future student/class records	Sensitive fields may require special permissions.

Role	Main powers	Sensitive restrictions
Teacher	Assigned class, attendance and permitted academic assessments	No school-wide student database by default.
Finance Officer	School fees, invoices, finance reporting; optional delegated billing view	No automatic account ownership or tender award.
Recruitment / HR Officer	Vacancies, candidates, staff records and onboarding	Restricted candidate info; no full owner authority.
Procurement Officer	Tenders, supplier bids and contracts	Sealed-bid access controlled; award may require two-person approval.
Marketing Officer	Public profile/media/promotions and approved statistics	No full admissions or child data.
Read-only Auditor	Scoped audit and reports	Cannot change operational records.
Super Admin Subroles	Verification, billing, moderation, support, security, auditor	Least privilege; case-by-case access to sensitive school data.

17.1 Staff screen and seat logic

Staff list: name, contact, role(s), invitation state, membership status, last activity and actions. Controls: invite, resend invitation, assign/edit role, suspend, revoke, review permissions, view history. Validate seat quota including proprietor and extras; invitation token expires and is single-use; owner change or staff departure revokes access promptly. Permissions are fine-grained read/create/edit/review/approve/publish/export/admin, not a simple unrestricted “manage” toggle. For high-risk finance, admission, ownership and tender actions provide approval separation where justified.

18. Super Admin Portal and platform governance

Admin area	Functions
Schools	Search, approve, request info, reject, suspend, archive, resolve duplicates and owner claims.
Verification	Review documents/identity, manage checklists, assign reviewers, record source and appeals.
Subscription & Billing	Configure plans/prices, discounts, credits, invoices, past-due states and approved refunds.
Quotas & Release Flags	Adjust seats, media, vacancies, tenders, region availability and unreleased features with audit.
Directory Moderation	Review profile revisions, media rights, public claims and badges.
Recruitment Moderation	Complaints, deceptive vacancies and publication policy enforcement.
Procurement Moderation	Tender disputes, supplier abuse, false documents and conflicting offers.
Regional Rollout	Enable school onboarding by Ghana region/district; not hardcoded to Greater Accra.
Reports	School registrations, conversion, billing, retention, support and aggregated metrics.

Admin area	Functions
Support	Tickets, appeals, authorised support session and escalation.
Security & Audit	Alerts, audit history, privileged actions, sensitive data-access records.

Super Admin does not have blanket unlimited routine access to student/admissions content. Structure platform owner, verification reviewer, billing administrator, moderation officer, support agent, security/ auditor with narrowly scoped permissions. Ownership transfer, high-value refund, sensitive exports, admin overrides and other high-risk actions require reason, elevated permission, sometimes dual approval, and immutable/tamper-evident audit record.

19. User experience and Theme B visual system

Reuse the exact Parents Portal v4.2 Theme B tokens; earlier discussions describe navy, royal blue, gold and white with Plus Jakarta Sans for headings and Inter for body/data. Do NOT invent precise HEX values when they have not been supplied in this task: retrieve from approved v4.2 document/source. The proprietor workspace is more information-dense than the parent discovery UI yet maintains shared brand and component consistency.

Design system category	Required specification
Layout	Desktop left sidebar; responsive compact mobile menu, top header, grid and spacing tokens inherited from Theme B.
Cards and tables	Readable neutral surfaces, modest borders, responsive data lists, pagination, meaningful empty states.
Inputs and forms	Visible labels, helper text, validations, error summary, save drafts, field grouping and appropriate auto-complete.
Buttons	Consistent primary/secondary/destructive hierarchy; confirmations for consequential actions.
Status	Use text+icon+accessible colour, never colour alone. Clear draft/pending/approved/expired definitions.
Photography	Use consented school photos, rights metadata, captions/alt text; protect children’s images.
Accessibility	WCAG 2.2 AA target, keyboard operation, contrast, focus states, screen-reader semantics, alternate chart table.
Performance	Optimised for lower-bandwidth Ghana mobile use; lazy-loading, small payloads, graceful failure.
System states	Loading/empty/success/error/no access/not entitled/pending verification/past due/unreleased.
Notifications	Actionable, scoped, dismiss/read semantics, channel preferences and history.

20. Technical architecture and source-of-truth contracts

Proposed modular architecture: existing frontend and API (verify actual code), shared authentication, Supabase PostgreSQL and secure object storage (verify), domain-based backend services, transactional

data changes, durable notification/events, analytics and monitoring. An independently deployed microservice for every module is unnecessary at launch. All tooling decisions must be justified against real repository evidence. Never create a second authentication provider or database by default.

Layer	Candidate / role	Audit requirement
UI	Existing JavaScript/React or other current framework	Inspect actual routes, components, code structure, Theme B usage.
Authentication	Existing supported provider	Check account reuse, sessions, MFA and guest flows.
Authorisation	Backend RBAC + resource checks + DB RLS	Inspect policies, service role leakage, tests.
Database	Existing Supabase PostgreSQL where suitable	Tables, relationships, unique constraints, indices, migrations, ownership.
Object storage	Private school-sensitive buckets and controlled public media	File classification, signed URLs, malware check, retention.
Backend	Modular APIs/business services	Validate request schema, transaction, idempotency, permission and error handling.
Payments	Verified Ghana payment provider such as Paystack	Reconfirm API features, recurring channels, fees and secure webhooks.
Notifications	Email, internal, SMS optional	Queue, retry, dedupe, preference and template history.
Automation	Backend jobs; optional n8n	Business-critical state not delegated to untrusted workflow output.
Analytics	Tenant-scoped aggregate reporting	Metric definitions, data minimisation, export scoping.
Operations	Staging, monitoring, backups, restore testing	Inspect hosting/deployment, secrets, audit and incident procedures.

21. Conceptual entity and relationships specification

Domain	Core records	Key relations and constraints
Identity	user, profile, session, invite	One identity multi-role; invites expire; membership explicit.
Institutions	school, education_level, school_location, campus_future	Single canonical school_id; duplicate match and legal name history.
Ownership	school_membership, school_claim, owner_transfer, role, permission	Unique active primary owner per school; owner one-school limit at launch.
Verification	verification_case, evidence_document, review_action	Reviewer identity, reasons, document-specific expiry and audit.
Directory	school_profile_revision, public_media, facility, programme	Published version separate from draft; rights and moderation.
Enquiries	enquiry, message, follow_up, assignment	One enquiry scoped to one school; source metadata and history.

Domain	Core records	Key relations and constraints
Admissions	intake, capacity, form_definition, form_version, applicant, guardian_contact, application, answer, document, assessment, offer, status_history	Submission preserves form version; application scoped to school/intake; authorised transitions.
Subscriptions	plan, plan_price, subscription, invoice, payment_transaction, refund_adjustment	Separate interval/prices; verified payment; historical snapshots.
Entitlements	feature_catalog, entitlement, usage_ledger, purchased_credit, credit_consumption	Atomic quota accounting and expiration handling.
Recruitment	vacancy, vacancy_revision, job_application, candidate_document, interview	Published state/slot usage and candidate scope.
Procurement	supplier, tender, tender_revision, supplier_bid, evaluation, contract	Confidential bid access and versioned publication.
Notifications	event_outbox, notification, template, delivery_attempt, preference	Durable events and retries without duplicate user effects.
Audit	audit_event, security_event, export_event	Actor, school, resource, action, time, reason and outcome.
Future SIS	student, verified_guardian_link, class, attendance, assessment, term, fee_invoice, fee_payment	Separate educational operations; link from accepted applications.

21.1 Authoritative record rules

- The schools domain owns school identity; the directory owns only moderated public revisions.
- Admissions owns application state; Parent Portal tracking and School Owner pipeline read the same authoritative state.
- Plan catalogue and price version represent commercial offers; individual subscription holds billing state and an entitlement snapshot/association.
- A bid belongs to one school-owned tender; vendor may respond to multiple tenders but cannot access competing bids.
- A parent may submit to multiple schools but each receives only the application explicitly submitted to it.
- Service events include stable IDs and references, avoiding unnecessary personal data payload.
- Admitted applicant to student conversion is idempotent; never create second student on a repeated enrolment event.

22. Backend operation/API contract catalogue

Paths are not fixed here. Antigravity must document actions by business semantics and fit them to existing API conventions; do not copy sample URL patterns blindly.

Operation group	Required actions	Critical controls
Institution	register-school; search-existing; claim-school; update-profile; publish-profile	Duplicates, approval, moderation, identity continuity.

Operation group	Required actions	Critical controls
Verification	submit-evidence; request-more; approve/ reject/escalate; transfer-owner	Reviewer permission, evidence access, audit.
School membership	invite; accept; assign-role; suspend/ revoke	Seats, one-school owner limit, single-use invite and scopes.
Enquiries	create-guest-enquiry; list; view; reply; assign; schedule; close/reopen	Rate-limit, verified contact, school tenancy and history.
Admissions	create-intake/form; publish-form; submit; review; request-doc; schedule; issue-offer; accept; enrol	Form version, status machine, guardian authority, capacity and idempotency.
Events/tours	create-slot; reserve; reschedule; cancel; record-attendance	Capacity and timezone consistency.
Billing	quote-price; start-checkout; verify-payment; change-plan; renew/cancel; list-invoices	Server trust, tax disclosure, idempotency and audit.
Entitlements	check-feature; reserve-quota; consume-credit; release/quota-reset	Atomic concurrency and consumption ledger.
Recruitment	draft/publish/pause/close/reopen vacancy; apply; review candidate	Active slot, privacy and deadlines.
Tenders	draft/publish/amend/close; submit-bid; list permitted bid; evaluate/award	Credit use, bidder confidentiality and opening policy.
SIS future	create/link-student; assign-class; attendance; results; fees	Student privacy, role scope, corrections and reconciliation.
Exports / reports	request-filtered-report; generate; retrieve	Authorised fields, rate-limits, signed short-lived links and export audit.

Each API contract requires: authenticated actor; target school/resource; request and response schema; validation; permissions; feature-release state; subscription entitlement; workflow state; concurrency rule; idempotency key if needed; rate limit; error codes; audit event; notification event; retry and observability behaviour. Never accept arbitrary client-supplied status or school_id as authority.

23. Cross-portal integration and event contract

Source	Target	Event / data flow	Access rule
Public Directory	School Owner	Approved profile and source metrics	Public published version only.
Parents Portal	School enquiries	Guest enquiry submitted; create school inbox item	School tenant, rate limiting, privacy.
Parents Portal	Admissions	Parent application and document upload	School/intake/version; confidential.
School admissions	Parents Portal	Information request, offer, decision, enrolment status	Verified parent/guardian only.
School profile	Directory and Matchmaker	Published revision + search update	Moderation and consistent school_id.

Source	Target	Event / data flow	Access rule
School recruitment	Teachers/candidates	Vacancy published, amended, closed	Correct public/job visibility and deadlines.
Candidate application	School recruitment	Candidate submission and status	Only recruiting school/ authorised HR.
School procurement	Suppliers Portal	Tender published and amended	Public/invited eligibility.
Supplier	School procurement	Bid and clarification	Bid privacy and opening controls.
Billing and verification	Super Admin	Review queue, subscription exceptions, complaints	Platform roles, audit and data minimisation.

23.1 Named event catalogue and delivery guarantees

- SchoolApproved, SchoolProfilePublished, ParentEnquirySubmitted, AdmissionApplicationSubmitted, AdmissionInformationRequested, AdmissionOfferIssued, AdmissionOfferAccepted, AdmissionEnrolmentConfirmed.
- VacancyPublished, VacancyClosed, JobApplicationSubmitted, TenderPublished, TenderAmended, SupplierBidSubmitted.
- SubscriptionActivated, SubscriptionRenewalFailed, SubscriptionExpired, PaymentVerified, CreditGranted, CreditConsumed, DocumentReviewRequired.
- Events should use unique event IDs, causal resource IDs, version, timestamp and minimal payload; consumers may receive duplicates and out-of-order messages.
- Use transactional outbox or equivalent durable event design for critical state changes; avoid sending a notification instead of committing a real business transaction.
- Enquiries, applications and payments require stable and consistent status when one notification channel is unavailable.

24. Security, privacy, child safeguarding and compliance

Handle children’s and guardian data as a high-risk domain. Ghana Data Protection Act 2012 (Act 843) and applicable data-protection guidance require a deliberate privacy programme, including determination of controller/processor roles by workflow, notices, lawful grounds, data minimisation, access management, retention and response to individual rights requests. Ghana regulatory evidence and official school licensing must be verified with current authorised guidance; this blueprint is not legal certification.

Control	Non-negotiable design requirement
Authentication	Secure login, recovery, MFA for owner/privileged accounts, session expiry, revoked invitations.
Tenant isolation	Enforce school-scoped queries at backend/DB; validate RLS and service-role usage.
Resource permissions	Teacher class scope, admissions role, supplier bid scope, owner vs staff, verified guardian links.
Storage	Private student/applicant/candidate/verification documents, file validation, malware scanning and controlled signed access.
Data protection	Purposes, notices, permissions, processor arrangements, cross-border transfer and retention decisions.

Control	Non-negotiable design requirement
Safeguarding	Need-to-know handling of sensitive cases, protected child images, privacy impact assessment.
Audit	Tamper-evident sensitive event logs, export logs, reason for admin overrides and retention policy.
Abuse resistance	Spam/rate limiting, secure inputs, anti-enumeration, duplicate fraud checks and upload limits.
Operations	Secrets management, monitoring, tested backups, incident response and restore playbook.
High-stakes decisions	Human decision-maker for admission denial, recruitment decisions and tender awards; AI may assist but not independently decide.

24.1 Negative tests that must fail

- School A user changes route ID/API school_id to access school B application, student, tender or billing data.
- Marketing officer uses global search to retrieve applicant documents or fee ledger.
- Teacher attempts to browse every student outside assigned class.
- Former owner or suspended employee performs an action with a retained session.
- Supplier obtains competing bid before allowed opening.
- Growth school uses direct API to access unreleased SIS or publish a third active vacancy.
- Repeated webhook or concurrent tender publish grants/consumes more credits than intended.
- Unauthorised guest guesses application tracking token and sees private child information.
- A public directory endpoint returns unpublished profile revisions or private documents.

25. Ghana payments, taxes and operational policies

The commercial schedule is approved, but tax display, SMS unit prices, advertising duration, purchased-credit expiry, vacancy repost rules, refunds and cancellation, grace-period length and school-level legal evidence checklist remain PENDING. Provider features and rates are VERIFY. Historical discussion referenced Paystack, card subscriptions and Mobile Money payment options; confirm currently supported recurring channels and fees from provider documentation before selecting a payment flow. No assumption of automatic recurring Mobile Money should enter production.

Pending ID	Unapproved policy	Recommendation for review (NOT approved)
P-01	Applicable Ghana taxes/levies	Confirm legal treatment, tax-inclusive/ exclusive display and invoicing.
P-02	SMS rate and credit packs	Test provider, publish per-message cost and delivery terms.
P-03	Featured advertising	Define 7/30-day plans, placement rights and clearly labelled promotion.
P-04	Purchased tender-credit expiry	Historical suggestion: 12 months; validate fairness/accounting and approve explicitly.

Pending ID	Unapproved policy	Recommendation for review (NOT approved)
P-05	Vacancy extension/reposting	Potential paid new 30-day period; define pause/close/repost effect.
P-06	Subscription grace period	Historical suggestion: 7 days; final approve timing and access rights.
P-07	Refund/cancellation terms	Published policy by product, statutory rights, proration and disputes.
P-08	Free enquiry limitations	Preserve basic receipt and response; define advanced CRM restriction.
P-09	Verification mandatory evidence	Align checklist to school level and applicable regulator.
P-10	Application fees and school funds	Do not enable until school/provider/legal/ merchant handling approved.
P-11	Paid pilot plan release	Sell Growth first; final pilot offer/ Enterprise future communications policy.
P-12	Reporting/retention values	Approval of privacy/data retention, export and post-cancellation windows.

26. Full phase-by-phase development roadmap (v1.6 preserved)

Phase	Scope and deliverables	Exit condition
Phase 0 — Audit & Foundation	Repository/frontend/backend/Supabase/ RLS/payment/deployment audit; architecture; gap report; migration proposal; access/security; test plan; rollback; change-control sign-off.	No implementation until codebase verified and approval recorded.
Phase 1 — Registration, Verification & Profile	Proprietor sign-up; new/claim school; dedupe; protected evidence; reviewer workflow; dashboard; public profile; staff invitation; audit.	Verified owner can manage one approved public school listing.
Phase 2 — Enquiry, Admissions & Billing MVP	Guest enquiry, inbox, follow-ups, intakes, versioned applications, tours, offers, parent tracking, approved plans, verified payments and entitlements.	A real parent-to-school application completes and paid Growth functionality works.
Phase 2B — Lightweight Marketplace	Vacancies, basic applications, tenders, supplier responses, quotas, purchases, credit ledger and publishing controls.	Published jobs/tenders accessible to intended candidates/suppliers; quota tests pass.
Phase 3 — SIS, Academics & Finance	Student/guardians, attendance, classes, assessments, results, enrolment migration, fees, receipts, school finance.	Private student and finance workflows secure and operational; Professional/ Enterprise unlock.
Phase 4 — Full HR & Procurement	Candidate review/interviews/offers, staff records/leave/qualifications, bid confidentiality/evaluation/awards/ contracts.	Complete process-based HR and purchasing with approved governance.

Phase	Scope and deliverables	Exit condition
Phase 5 — Enterprise & Intelligence	Analytics, optional AI, automation, integrations, inventory, facilities, future multi-school ownership.	Separate validation and authorisation; no assumption of initial availability.

26.1 Phase 0 audit packages — all mandatory

ID	Audit scope
AUD-01	Repository structure, Git history, dependencies, uncommitted work
AUD-02	Frontend and Theme B UI screens, responsiveness and accessibility
AUD-03	API framework and business operations
AUD-04	Supabase schema, relationships, indices, RLS, migrations
AUD-05	Authentication, owner/staff roles, sessions and service-role exposure
AUD-06	Payment and subscription/provider integrations
AUD-07	Cross-portal existing school/enquiry/application IDs and flows
AUD-08	Hosting, deployments, environments, secrets, logs, backups and recovery

26.2 Phase 1 detailed build packages

ID	Package
SCH-001	Proprietor registration and contact verification
SCH-002	Existing-school lookup and claim conflict prevention
SCH-003	Registration wizard with institution fields
SCH-004	Private verification document submission
SCH-005	Admin reviewer queue and decisions
SCH-006	Workspace activation after authority confirmed
SCH-007	School overview dashboard and tasks
SCH-008	Public profile draft/preview/moderation/publication
SCH-009	Staff invitation, membership, roles and seat check
SCH-010	Audit history and action traceability

26.3 Phase 2 detailed build packages

ID	Enquiry / admissions / billing item
ENQ-001	Guest enquiry form and abuse protection

ID	Enquiry / admissions / billing item
ENQ-002	Enquiry routing to authoritative school
ENQ-003	School inbox and scoped lists
ENQ-004	Conversation history and reply
ENQ-005	Assignment and follow-up tasks
ENQ-006	Enquiry status and closure/reopening
ENQ-007	Email/in-platform notifications
ENQ-008	Enquiry analytics and conversion
ADM-001	Intake and school academic year
ADM-002	Capacity configuration
ADM-003	Versioned form builder and custom questions
ADM-004	Parent application submission
ADM-005	Application review and staff assignment
ADM-006	Secure supporting documents
ADM-007	Valid admission state transitions
ADM-008	Tours, interviews and assessments
ADM-009	Offers, decline and waitlist
ADM-010	Private parent application tracking
ADM-011	Admissions reporting and export
ADM-012	Enrolment confirmation without SIS
SUB-001	Plan and approved price catalogue
SUB-002	Monthly/annual billing intervals
SUB-003	Checkout and authorisation flow
SUB-004	Verified transaction reconciliation
SUB-005	Feature entitlement engine
SUB-006	Seat/media and future vacancy/tender quota design
SUB-007	Upgrade/downgrade impacts
SUB-008	Renewal/past due/grace/expiry states
SUB-009	Billing receipts and invoices
SUB-010	Super Admin commercial settings

27. Greater Accra pilot and national expansion

Proposed pilot cohort: 10-20 private schools across Accra, Tema and surrounding Greater Accra communities, mixing nursery/KG, primary, JHS and where applicable SHS, small and medium schools and a few larger institutions. This number is a target rather than a claimed recruited cohort. Do not hardcode Greater Accra into database schema; use country > region > district/municipality > locality > verified geolocation and configurable onboarding region flags. Public school browsing may cover other Ghanaian regions even while school-owner paid onboarding is restricted.

Pilot gate	Definition / evidence
Internal alpha	Authentication, RLS, registration, school profile and basic workflows tested with safe test data.
Private beta	Selected school owners complete claim/onboarding and publish accurate profiles.
Admissions pilot	Real consenting parents and schools complete enquiry, tour and application workflows reliably.
Paid pilot	Plan checkout, fees, renewals, quota, failed payments and reporting validated.
Regional launch	Support, security, accessibility, incident/recovery and commercial policies ready.
National scale	Retention, usage, satisfaction, conversion and costs support expansion.

27.1 Pilot success metrics — targets, not observed results

Metric	Initial success hypothesis
Invited school onboarding completion	At least 80% of pilot schools, target subject to actual recruitment.
Approved school profile completion	At least 80% of participating schools.
Accepted enquiry recording	At least 99% correct persistence/routing; no silent drop.
Application state consistency	100% of tested valid transitions.
Cross-school security	Zero successful unauthorised access in acceptance/security testing.
Payment reconciliation	100% of tested completed transactions correctly reconciled.
Accessibility blockers	Zero known critical launch blockers.
Business validation	Interviews with approx. 15-20 proprietors, price sensitivity around GH¢69/99/149 Growth test points.
Commercial metrics	Weekly active administrators, lead response, application conversion, paying retention, revenue per school, support load and gross margin.

28. Functional, integration, security and recovery testing

28.1 E2E acceptance catalogue from v1.6

ID	Scenario	Expected result
E2E-01	New owner registers school	Pending application with correct institution identity.
E2E-02	Existing listing claimed	No duplicate active owner; evidence review.
E2E-03	Reviewer approves school	Authorised school workspace activated.
E2E-04	Owner publishes profile	Approved version visible publicly, drafts private.
E2E-05	Guest parent submits enquiry	Correct school inbox receives record; no forced login.
E2E-06	Parent applies for admission	Correct school/intake/form version saved.
E2E-07	School makes admission decision	Valid state transition and authorised parent tracking updated.
E2E-08	Growth subscription paid	Only verified successful payment activates proper entitlements.
E2E-09	Membership expires	Premium write restrictions without data loss.
E2E-10	School exceeds vacancy quota	Publication denied even via backend request.
E2E-11	Tender credit purchased	Exactly one ledger credit for one verified transaction.
E2E-12	Staff departs / revoked	All further private operations denied.
E2E-13	School A requests School B data	Denied and logged.
E2E-14	Deployment interrupted	Data integrity and tested recovery/ rollback.

28.2 Edge case and resilience catalogue

- Concurrent edits on one application: version conflict and explicit merge/retry.
- Repeated guest enquiry: spam defence plus safe deduplication preserving valid messages.
- School tour booked by two people at last capacity: only allowed remaining slots are reserved.
- Owner approval and staff invitation racing with seat count: authoritative enforcement.
- Subscription expired while user holds a draft: preserve draft, restrict premium final action.
- Profile unpublished: public display withdrawn, workspace retained.
- Simultaneous vacancy publication or tender-credit consumption: no overspending.
- Payment webhook repeated/out of order: idempotent verified state and reconciliation.
- Tender closes while bidder uploads: deadline enforced reliably.
- Unsafe file uploaded: reject/quarantine and notify with safe error.
- Child/guardian link guessed: no private access granted.

- Failed email/SMS delivery: committed record persists; retry/outbox and operator visibility.
- Requested export: field-scope privacy check, signed link, expiration, audit.

29. Definition of done and approval gates

- A requirement has a stable ID, role, plan, release phase, precise expected behaviour and acceptance criteria.
- Each implemented view uses Theme B; valid states for loading, empty, permission-denied, upgrade-required, unreleased, errors and success.
- Backend operation and database write genuinely work; never claim a mock screen is complete.
- RLS/permission/entitlement validated with negative tests and appropriate audit records.
- Cross-portal event and related parent/school views tested against one authoritative school record.
- Regression tests demonstrate existing Parents Portal behaviour was not broken.
- Payment and quotas tested for duplicate/cross-event and concurrency edge cases.
- Code reviewed, migrations tested in staging, backup and rollback reviewed.
- Phase approval recorded before deploying to production; release evidence and known issues attached.

Gate	Approval evidence
Gate 0 Audit	Real repo/database inspection, source evidence, gap register, risk assessment, access and environment plan.
Gate 1 Architecture	Canonical data model, RLS strategy, API contract, migration/backup proposal.
Gate 2 Product	MVP screens/forms/workflows/roles, no false future feature claims.
Gate 3 Commercial	Approved prices/quotas plus pending tax/refund/credit rules resolved for live checkout.
Gate 4 Privacy/regulatory	Relevant legal review, school licensing evidence checklist, child-data privacy impact assessment.
Gate 5 Staging acceptance	Integrated functional/security/accessibility/payment/recovery test evidence.
Gate 6 Pilot release	Support plan, incident response, controlled Greater Accra onboarding and monitoring.

30. Major risks and mitigation register

Risk	Severity	Mitigation
Cross-school private data leak	CRITICAL	Tenant-aware API + RLS + negative tests + restricted service credentials.
Child/guardian information exposure	CRITICAL	Private storage, verified guardian links, minimisation and DPIA.
Existing schema incompatible	HIGH	Audit first, versioned migrations, backfill and rollback review.
Duplicate school/owner claim	HIGH	Canonical match, one owner uniqueness, reviewer dispute process.

Risk	Severity	Mitigation
Parent-school application divergence	HIGH	Single admissions source of truth and event reconciliation.
Double subscription/tender credit	HIGH	Verified payment reference, idempotency, atomic ledger.
Paid features bypassed via API	HIGH	Backend role/plan/release validation and RLS.
Too many launch modules	HIGH	Strict Phase 1 + Phase 2 MVP scope; later releases feature-flagged.
Unclear Ghana licensing evidence	HIGH	School-type policy and regulator verification; no false badges.
Paid conversion below plan	HIGH	Greater Accra interviews, pilot and willingness-to-pay tests.
Costs exceed small subscriptions	MEDIUM/HIGH	Measure SMS, media, support, hosting, gateway and gross margin.
Existing Parents Portal breaks	HIGH	Dependency mapping, regression testing and staged release.
Owner transfer creates access ambiguity	HIGH	Atomic revoke/grant and case review with audit.
Feature promises exceed delivered code	HIGH	Release availability independent of entitlement; public clear labelling.

31. Innovation backlog preserved from v1.0

Innovation	Detailed future value / safety
School Growth Intelligence	Admissions sources, enquiry response, bottlenecks, capacity, forecasts; recommendations grounded in real data.
Permission-aware AI assistant	Draft replies, cite sources, report explanation, suggest tasks; require confirmation on changes and prohibit autonomous rejection/hiring/award.
Events and open days	RSVP, registration capacity, attendance, reminder and follow-up lead link.
Parent-school communication hub	Private child-specific updates, notices and meeting scheduling after guardian verification.
Compliance and expiry centre	Document/certificate reminders, school readiness checklists without claims of automatic licensing compliance.
Admissions capacity forecasting	Estimated places, active applications and projected intake with uncertainty shown.
Low-bandwidth/mobile support	Fast touch-friendly app, drafts and carefully defined offline-friendly workflows.
School Group HQ	Later multi-campus consolidation, delegated administrators, subscriptions and reporting.

Innovation	Detailed future value / safety
Other extensions	Assets/facilities, maintenance tickets, inventory, school transport, meals, library, alumni, surveys and configurable forms.

32. Detailed requirements traceability: representative high-risk register

Requirement ID	Description	Authority	Phase	Verification
REQ-PUB-001	Public school browse/ search/compare without login	Parents v4.2 approved	Existing + 1/2	Guest E2E test
REQ-OWN-001	One active primary proprietor-school workspace	v1.1 approved	1	Unique/role negative tests
REQ-OWN-002	Delegated authorised school staff	v1.0/1.1 approved	1	Invitation and seat tests
REQ-COM-001	GH¢ prices and annual discount	v1.5 APPROVED	2	Invoice/pricing acceptance
REQ-COM-002	User/media/vacancy/ tender quotas	v1.5 APPROVED	1/2/2B	Entitlement tests
REQ-COM-003	Approved add-on prices	v1.5 APPROVED	2B	Payment/credit ledger tests
REQ-VER-001	Owner/regulatory/ subscription separate statuses	v1.2 baseline	1	Reviewer case walkthrough
REQ-ADM-001	School-scoped versioned application	v1.2-1.4 baseline	2	Versioning and privacy E2E
REQ-ADM-002	Human-authorised state transitions	v1.4 baseline	2	Illegal status transition test
REQ-ADM-003	Enrolment confirmation without SIS	v1.1 approved	2	Admission successful E2E
REQ-JOB-001	2 active Growth jobs and extra slot GH¢25/30d	v1.5 APPROVED	2B	Quota & payment tests
REQ-TEN-001	3/10 monthly higher-tier tenders + credits	v1.5 APPROVED	2B	Concurrent publication test
REQ-SIS-001	Student/academic/fees Professional+ when released	v1.5 APPROVED	3	Feature flag and privacy tests
REQ-SEC-001	No cross-school operational access	v1.0-1.6 critical	0 onward	RLS/API penetration tests
REQ-SYS-001	Audit actual repo and Supabase before coding	v1.6 binding process	0	Source-evidenced audit

Requirement ID	Description	Authority	Phase	Verification
REQ-GEO-001	Greater Accra pilot, nationwide-ready data	v1.1 approved	1/2	Region configuration review

Antigravity must expand this representative register to EVERY screen, field, action, state and test in its Phase 0 planning deliverables. “Representative” does not authorise dropping the detailed requirements elsewhere in this document. All historical detail above is in scope for design; only the appropriate phase is in scope for construction.

33. Pending decisions and assumptions register

Do not treat these recommendations as approved: applicable taxes; SMS unit price; featured promotion duration/placement; tender-credit validity; vacancy renewal or repost conditions; grace-period length (seven days was suggested, not approved); refund/cancellation terms; basic Free enquiry restrictions; exact institution evidence by school category; admission application fee policy; data retention durations; final pilot recruitment participants; detailed technical framework choices; production provider support/rates. Confirm with product owner, legal/regulatory advisers and source evidence as appropriate.

34. Source verification and external dependencies

Prior planning mentioned NaSIA pre-tertiary licensing guidance, Ghana Data Protection Act 2012 (Act 843), Paystack Ghana payment options and local Ghana school management competitors. Those references informed strategy but are not current legal certification or an independently performed v2.0 web research audit. Before coding and live sale, Antigravity or the project team must verify official regulatory requirements, payment capabilities/fees, tax treatment and competitor pricing at the then-current date, preserving URL/source, access date and evidence. Never silently incorporate stale claims as fixed technical requirements.

35. Master developer handoff and precedence rules

- Give Antigravity this v2.0 PDF AND the separate Master Implementation Prompt; no need to paste prior v1.0-v1.6 messages if this PDF is attached and confirmed readable.
- Ask Antigravity to inventory real source and report gap, not to build immediately.
- Where live code differs from the blueprint, document the difference and proposed non-destructive compatibility plan; obtain a decision rather than silently overriding commercial rules.
- Expand every detailed module into implementable user stories and acceptance tests and map to existing routes/tables.
- Phase 0 output is an architecture assessment, not a production deployment.
- Promote Phase 1 only after architecture, backup, privacy and security plans reviewed and explicit implementation scope authorised.

36. Full audit-first deliverables list

- D01 Existing architecture/codebase audit with direct evidence.
- D02 Current-feature inventory and gap matrix (existing/partial/missing/defective/conflicting/unverified/deferred).
- D03 Consolidated Product Requirements Document with traceability to every section of v2.0.

- D04 Conceptual + logical ERD, entity dictionary, relationships and uniqueness constraints.
- D05 Permission matrix, tenant isolation and proposed RLS review.
- D06 Full sidebar, screens, buttons, fields, states and responsive UX specification.
- D07 Admissions form/version and state transition specification.
- D08 Enquiry and tour workflows and cross-portal data contracts.
- D09 Subscription, price, entitlement, quota, add-on, invoice and payment architecture.
- D10 Vacancy/tender publishing, candidate/supplier response and credit logic.
- D11 Notification/event catalogue and idempotent delivery design.
- D12 API operation catalogue with auth, failure and audit contracts.
- D13 Security, data protection, retention and safeguarding plan.
- D14 Supabase migrations/backfill/backup/rollback proposal (planning only).
- D15 Theme B source token/component parity audit.
- D16 Phase 1 and Phase 2 implementation-ready tasks and dependencies.
- D17 Phase 2B-5 scoping and feature-release roadmap.
- D18 Test plan with E2E, negative security, concurrency, accessibility and recovery.
- D19 Pilot operations, support and measurement plan.
- D20 Operating cost, SMS/media/payment unit economics and risks.
- D21 Open decision log and approval requests.
- D22 Implementation go/no-go decision report.

37. Final v2.0 acceptance and handoff checklist

Acceptance item	Expected status / requirement
All historical versions preserved in substance	Sections 1-36 include v1.0-v1.6 module, screen, commercial, architecture and test details.
Approved commercial configuration exact	GH¢ prices, annual discount, quotas and add-ons reflect explicit approval.
Pending items labelled	Tax, SMS price, marketing duration, credits, grace, refunds, legal checklist not silently fixed.
Launch vs future separated	Registration/admissions/commerce launch; jobs/tenders 2B; SIS/fees 3; HR/procurement 4; enterprise 5.
Parents Portal public access preserved	No mandatory login to browse/search/compare schools.
Security and child-data protections explicit	Tenant isolation, verified guardians, audited access and controlled files.
Technical claims honestly represented	Codebase, Supabase and deployment not inspected in this document.
Antigravity scope constrained	Phase 0 planning/audit first; no source, database or production mutation.
Separate implementation prompt provided	Companion document contains precise ordered operational instructions.

END OF SKOOLFINDER GHANA SCHOOL OWNER & PROPRIETOR PORTAL MASTER BLUEPRINT v2.0

Document authority: Approved commercial and strategic decisions are frozen; implementation remains unauthorised. The companion Antigravity prompt governs the audit-first delivery sequence.